MTH107 2021S1 Exam

Feel free to contact me at maths_and_programming@outlook.sg if you need tuition. I provide university level tuition for the following SUSS units:

* ENG201 Linear Systems Analysis and Design
* ENG301 Microprocessor Programming
* ICT101 Computational Thinking
* ICT133 Structured Programming
* ICT162 Object Oriented Programming
* MTD215 Application of C++ in Multimedia
* ICT231 Data Structures and Algorithms I
* ICT232 Data Structures and Algorithms II
* MTH105 Fundamentals of Mathematics
* MTH107 Calculus I
* MTH108 Calculus II
* MTH316 Multivariable Calculus
* MTH319 Mathematical Methods I
* MTH320 Mathematical Methods II
* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference
* MTH355 Basic Mathematical Optimisation

Donaldson Tan

MTH107

Timed Online Assignment – January Semester 2021

Calculus I

Tuesday, 18 May 2021 **1:00 pm – 3:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FIVE (5)** questions and comprises **FOUR (4)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O)*.

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

(a) Determine the following limit. Show your steps clearly.

$$\lim_{x \to 1} \frac{x^7 - 1}{x - 1}$$

(6 marks)

(b) Determine the following limit. Show your steps clearly.

$$\lim_{x \to \infty} \frac{x + 3}{\sqrt{x^2 + 3}}$$

(8 marks)

(c) Determine whether the equation $\tan x = 1 - x$ has a real root in the interval $[0,1]$. Justify your answer.

(6 marks)

Question 2

(a) Let f be the function defined on $(-\infty, \infty)$ by

$$f(x) = \begin{cases} x^2 \sin(\frac{1}{x^2}) & if\ x \neq 0 \\ 0 & if\ x = 0 \end{cases}.$$

Determine whether f is continuous at 0.

(7 marks)

(b) Let f be the function defined by

$$f(x) = \begin{cases} \frac{x^3+3x}{|x|} & if\ x \neq 0 \\ a & if\ x = 0 \end{cases}.$$

Is it possible to choose a suitable value of a such that f is continuous at 0? Justify your answer.

(7 marks)

(c) Let $f(x) = x^2|x|$. Is f differentiable at 0? If yes, determine the value of $f'(0)$. Justify your answer.

(6 marks)

Question 3

(a) Let $f(x)$ be a function that is differentiable for all x. Let $g(x)$ be defined by $g(x) = f(x) + f(3 - x)$. Show that $g'(x)$ has a root in the interval $(0,3)$.

(10 marks)

(b) Let $f(x)$ be continuous and differentiable on the interval $[-8,0]$. Suppose $f(-8) = -2$ and $f'(x) \leq 3$ for all x. What is the largest possible value for $f(0)$? Justify your answer.

(10 marks)

Question 4

(a) Let $f(x) = e^{x^3}$. Determine the value of $f^{(21)}(0)$. You may leave your answer in terms of factorials, in the form $\frac{a!}{b!}$.

(10 marks)

(b) Determine the Taylor's expansion of $\sin 2x$ about $a = 0$, up to and including the term in x^3. Use this Taylor expansion to estimate the value of $\sin 0.2$, correct to 3 decimal places.

(10 marks)

Question 5

Let f be the function defined on $(-\infty, \infty)$ by

$$f(x) = \frac{x}{\sqrt{(1+x^2)^3}}.$$

(a) Apply the first derivative test to compute the interval on which f is increasing.

(5 marks)

(b) Determine the horizontal and vertical asymptotes of f if they exist. Justify your answers.

(5 marks)

(c) Determine the critical point(s) of f and discuss their nature. Show your steps clearly.

(5 marks)

(d) Sketch the graph of f.

(5 marks)

----- END OF PAPER -----

1 (a) $\lim_{x \to 1} \frac{x^7 - 1}{x - 1} = \lim_{x \to 1} \frac{(x-1)(x^6 + x^5 + x^4 + x^3 + x^2 + x + 1)}{x - 1}$ ← use difference of powers formula

$a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \dots + ab^{n-2} + b^{n-1})$

$= \lim_{x \to 1} (x^6 + x^5 + x^4 + x^3 + x^2 + x + 1)$

$= 7$

(b) $\lim_{x \to \infty} \frac{x+3}{\sqrt{x^2+3}} = \lim_{x \to \infty} \frac{\frac{1}{x}(x+3)}{\frac{1}{x}\sqrt{x^2+3}}$

$= \lim_{x \to \infty} \frac{1 + \frac{3}{x}}{\sqrt{1 + \frac{3}{x^2}}}$

$= \frac{1+0}{\sqrt{1+0}}$

$= 1$

(c) $\tan x = 1 - x$

$\tan x - 1 + x = 0$

let $f(x) = \tan x - 1 + x$

$f(0) = -1$

$f(1) = \tan 1$

By Intermediate Value Theorem,

Since $f(0) < 0$ and $f(1) > 0$,

$\exists c$ on $[0, 1]$ such that $f(c) = 0$

$\therefore$ There exists a real root on $[0, 1]$

2(a) $-1 \le \sin\left(\frac{1}{x^2}\right) \le 1$

$-x^2 \le x^2 \sin\left(\frac{1}{x^2}\right) \le x^2$

$\lim_{x\to 0} -x^2 \le \lim_{x\to 0} x^2 \sin\left(\frac{1}{x^2}\right) \le \lim_{x\to 0} x^2$

$0 \le \lim_{x\to 0} x^2 \sin\left(\frac{1}{x^2}\right) \le 0$

By Squeeze Theorem, $\lim_{x\to 0} x^2 \sin\left(\frac{1}{x^2}\right) = 0$

$\lim_{x\to 0} f(x) = \lim_{x\to 0} x^2 \sin\left(\frac{1}{x^2}\right) = 0$

$f(0) = 0$

Since $f(0) = \lim_{x\to 0} f(x)$, then $f(x)$ is continuous at $x=0$.

(b) $\lim_{x\to 0^+} f(x) = \lim_{x\to 0} \frac{x^3+3x}{|x|}$

$= \lim_{x\to 0} \frac{x^3+3x}{x}$

$= \lim_{x\to 0} (x^2+3)$

$= 3$

$\lim_{x\to 0^-} f(x) = \lim_{x\to 0} \frac{x^3+3x}{|x|}$

$= \lim_{x\to 0} \frac{x^3+3x}{-x}$

$= -\lim_{x\to 0} (x^2+3)$

$= -3$

$\therefore \lim_{x\to 0} f(x)$ doesn't exist.

$\therefore$ It is not possible to choose a value of a such that f is continuous at 0.

(c) $\lim_{h\to 0^+} \frac{f(h+0)-f(0)}{h} = \lim_{h\to 0^+} \frac{h^3-0}{h} = \lim_{h\to 0^+} h^2 = 0$

$\lim_{h\to 0^-} \frac{f(h+0)-f(0)}{h} = \lim_{h\to 0^-} \frac{h^2(-h)-0}{h} = \lim_{h\to 0^-} -h^2 = 0$

$\therefore \lim_{h\to 0} \frac{f(h+0)-f(0)}{h}$ exists and $\lim_{h\to 0} \frac{f(h+0)-f(0)}{h} = 0$

$\therefore f'(0) = \lim_{h\to 0} \frac{f(h+0)-f(0)}{h} = 0$

3(a) $g(x) = f(x) + f(3-x)$

By Mean Value Theorem, $\exists c$ on $(0,3)$ such that

$$g'(c) = \frac{g(3)-g(0)}{3-0}$$

$$= \frac{(f(3)+f(0)) - (f(0)+f(3))}{3}$$

$$= 0$$

$\therefore g'(x)$ has a root in $(0,3)$

(b) By Mean Value Theorem, $\exists c$ on $(-8,0)$ such that

$$f'(c) = \frac{f(0)-f(-8)}{0-(-8)}$$

$$= \frac{f(0)-(-2)}{8}$$

$$= \frac{f(0)+2}{8}$$

$$f'(x) \leq 3$$

$$\Rightarrow f'(c) \leq 3$$

$$\frac{f(0)+2}{8} \leq 3$$

$$f(0)+2 \leq 24$$

$$f(0) \leq 22$$

$\therefore \max f(0) = 22$

4(a) $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \Rightarrow e^{x^3} = \sum_{n=0}^{\infty} \frac{(x^3)^n}{n!} = \sum_{n=0}^{\infty} \frac{x^{3n}}{n!}$

Using Taylor Expansion,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} (x-0)^n$$

$$\Leftrightarrow \frac{f^{(21)}(0)}{21!} x^{21} = \frac{x^{3k}}{k!} \text{ where } \begin{array}{l} 3k = 21 \\ k = 7 \end{array} = \frac{x^{21}}{7!}$$

$$\therefore f^{(21)}(0) = \frac{21!}{7!}$$

(b) $\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$

$$\sin 2x = \sum_{n=0}^{\infty} (-1) \frac{(2x)^{2n+1}}{(2n+1)!} = 2x - \frac{8x^3}{3!} + \frac{32x^5}{5!} - \dots$$

$\sin 0.2 = \sin 2x$ where $x = \frac{0.2}{2} = 0.1$

$$\simeq 2x - \frac{8x^3}{3!}$$

$$= 0.1987$$

5(a) $f(x) = \dfrac{x}{\sqrt{(1+x^2)^3}} = x(1+x^2)^{-\frac{3}{2}}$

$$f'(x) = (1+x^2)^{-\frac{3}{2}} + x\left(-\tfrac{3}{2}\right)(1+x^2)^{-\frac{5}{2}}(2x)$$

$$= \frac{1}{\sqrt{(1+x^2)^3}} - \frac{3x^2}{\sqrt{(1+x^2)^5}}$$

$$= \frac{(1+x^2) - 3x^2}{\sqrt{(1+x^2)^5}}$$

$$= \frac{1-2x^2}{\sqrt{(1+x^2)^5}}$$

let $f'(x) > 0$

$1 - 2x^2 > 0$

$2x^2 - 1 < 0$

$x^2 < \frac{1}{2}$

$-\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$

$\therefore f(x)$ is increasing when $-\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$

(b) $x^2 \geq 0 \Rightarrow 1+x^2 \geq 1 \Rightarrow \sqrt{(1+x^2)^3} \geq 1$

$\therefore$ There is no vertical asymphote

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{x}{(1+x^2)^{\frac{3}{2}}}$$

L'Hopital Rule

$$= \lim_{x\to\infty} \frac{1}{\frac{3}{2}(1+x^2)^{\frac{1}{2}}2x}$$

$$= \lim_{x\to\infty} \frac{1}{3x\sqrt{1+x^2}}$$

$$= 0$$

$$\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty} \frac{x}{(1+x^2)^{\frac{3}{2}}}$$

L'Hopital Rule

$$= \lim_{x\to-\infty} \frac{1}{3x\sqrt{1+x^2}}$$

$$= 0$$

$\therefore y=0$ is the horizontal asymphote

5(c) let $f'(x) = 0$

$\Rightarrow 1 - 2x^2 = 0$

$x = \pm \frac{1}{\sqrt{2}}$

$f\left(\frac{1}{\sqrt{2}}\right) = \frac{2\sqrt{3}}{9}$

$f\left(-\frac{1}{\sqrt{2}}\right) = -\frac{2\sqrt{3}}{9}$

$$\cancel{f''(x) = \frac{d}{dx}\frac{x^2-3x+1}{\sqrt{(1+x^2)^5}}}$$

$$\cancel{= (2x-3)(1+x^2)^{-\frac{5}{2}} + (x^2-3x+1)\left(-\frac{5}{2}\right)(1+x^2)^{-\frac{7}{2}}(2x)}$$

$$\cancel{= \frac{2x-3}{\sqrt{(1+x^2)^5}} - \frac{5(x^2-3x+1)}{\sqrt{(1+x^2)^7}}}$$

$$f'(x) = \frac{d}{dx}(1-2x^2)(1+x^2)^{-\frac{5}{2}}$$

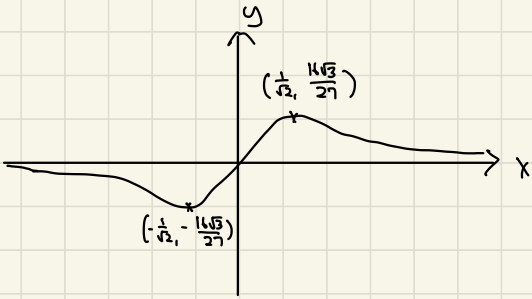
$$= -4x(1+x^2)^{-\frac{5}{2}} - \frac{5}{2}(2x)(1-2x^2)(1+x^2)^{-\frac{7}{2}}$$

$$= -\frac{4x}{\sqrt{(1+x^2)^5}} - \frac{5x(1-2x^2)}{\sqrt{(1+x^2)^7}}$$

$f''\left(\frac{1}{\sqrt{2}}\right) = -\frac{16\sqrt{3}}{27} < 0$

$\Rightarrow \left(\frac{1}{\sqrt{2}}, \frac{2\sqrt{3}}{9}\right)$ is a maxima

$f''\left(-\frac{1}{\sqrt{2}}\right) = \frac{16\sqrt{3}}{27} > 0$

$\Rightarrow \left(-\frac{1}{\sqrt{2}}, \frac{16\sqrt{3}}{27}\right)$ is a minima

(d)